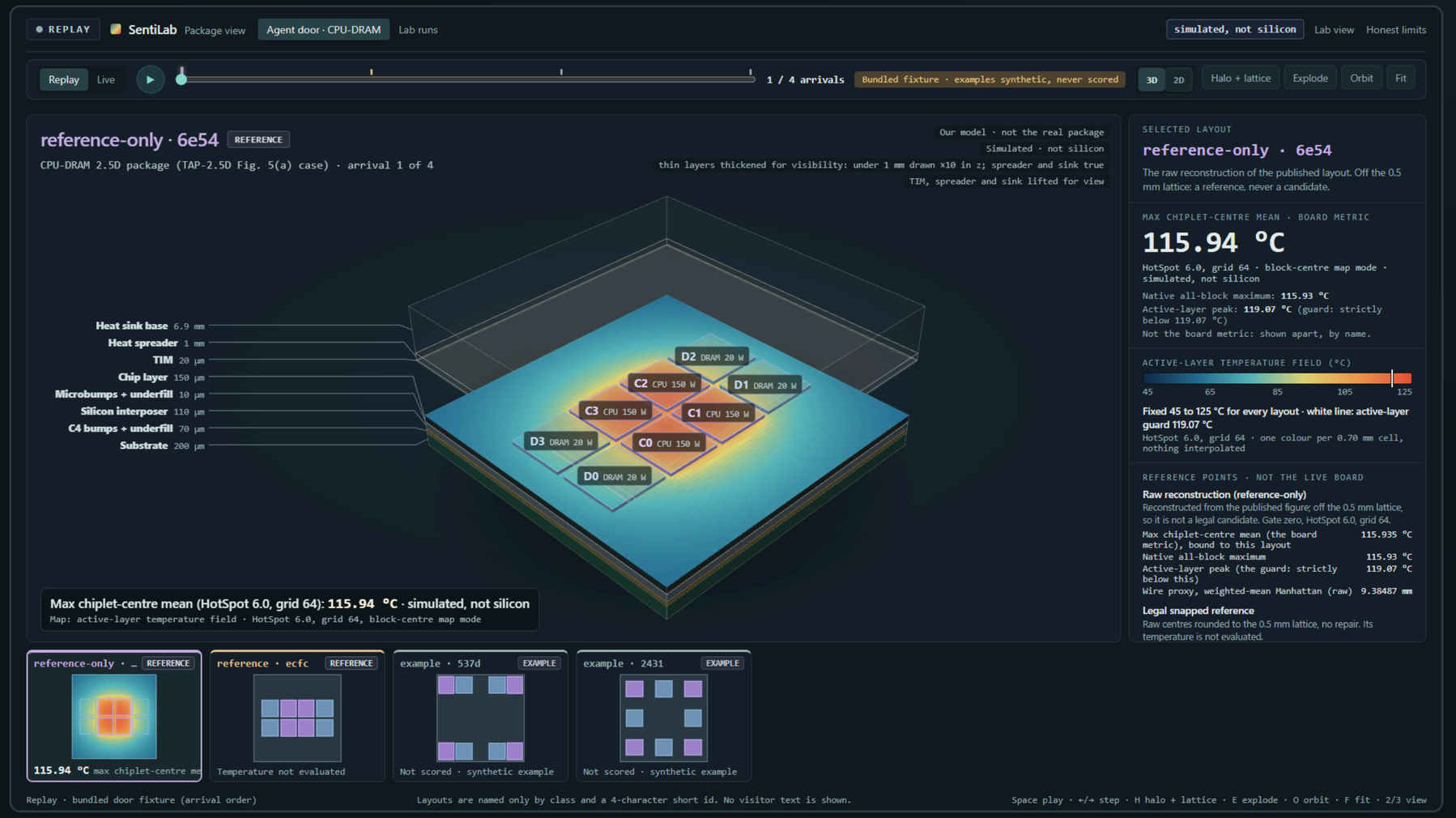
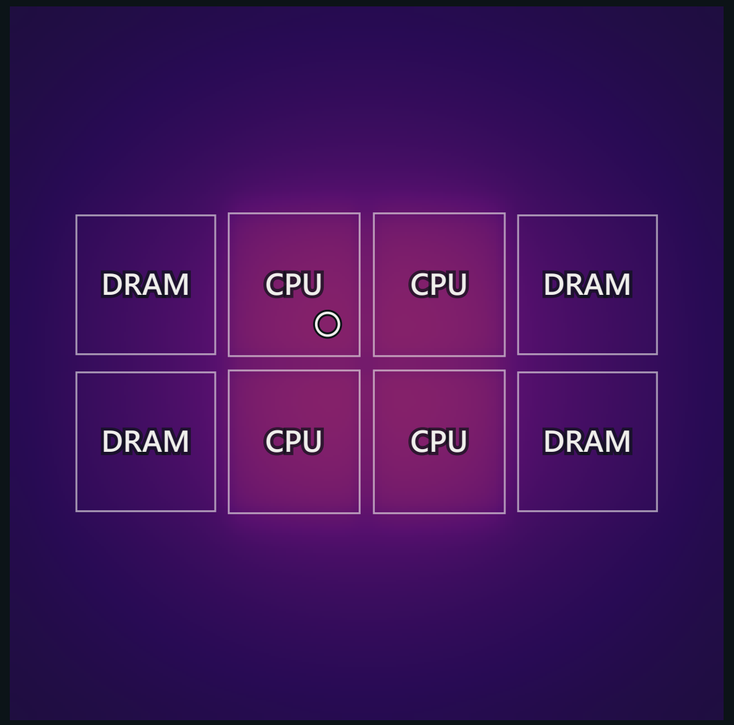
AI agents arrange chips. Software predicts the heat.

simulated, not silicon

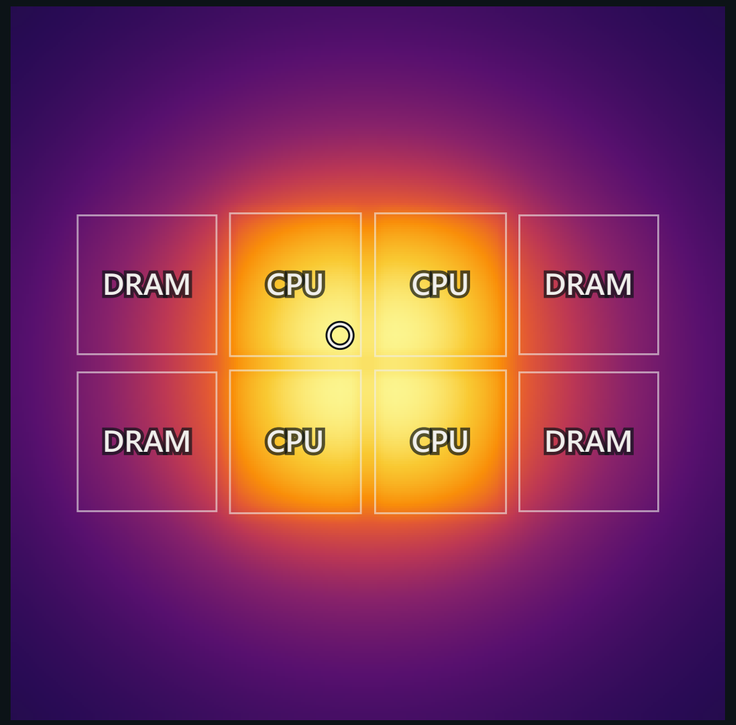


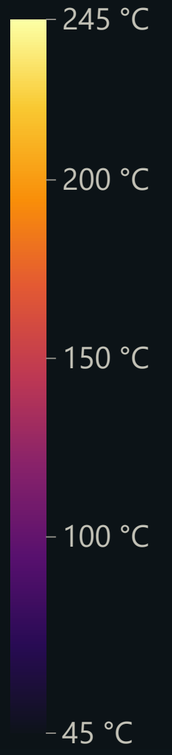
Same design, same files: 119 °C or 241 °C?

Older, 6.0 hottest spot 119 °C



Newer, 7.0 hottest spot 241 °C





2.6×

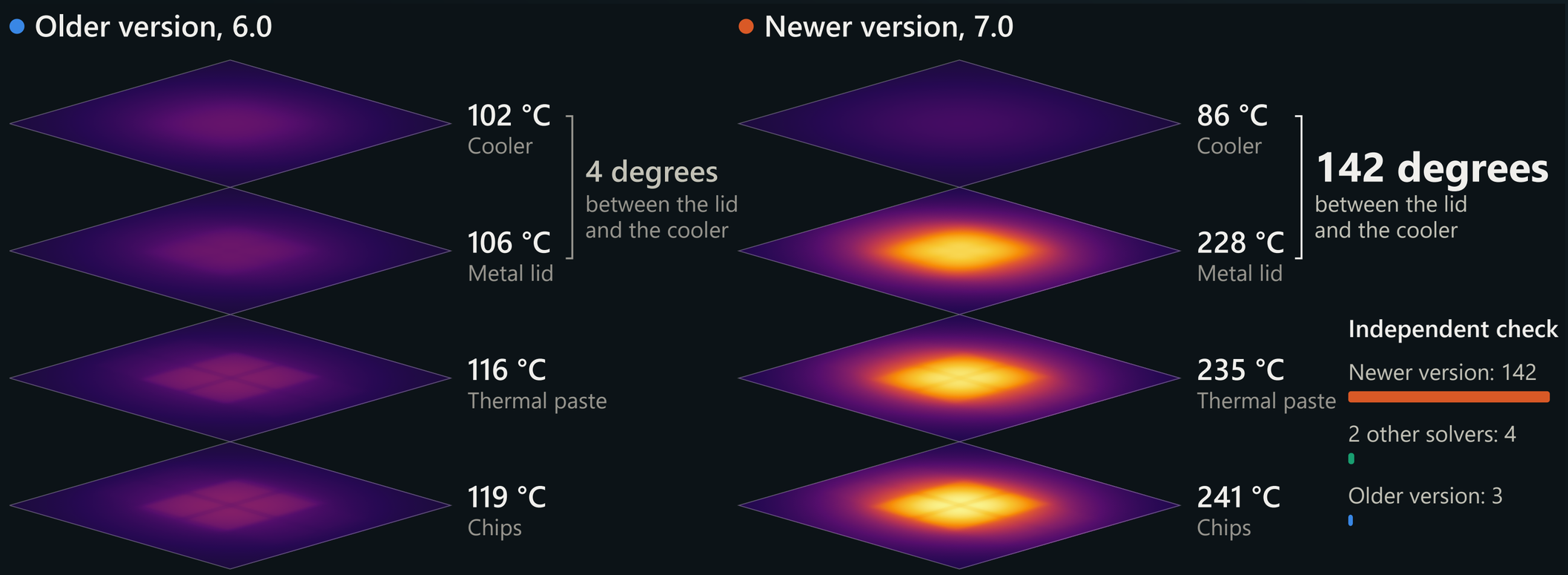
as much heating, above the 45 °C air

Same chips
Same power
Same input files

Facts as of 2026-11-01 03:50 UTC

The CPU-DRAM reference layout. Both versions run in the detailed-3D mode the published method uses. The ring marks the hottest point of the chip layer. Run 2026-10-31 23:41-23:44 UTC. Simulated, not silicon.

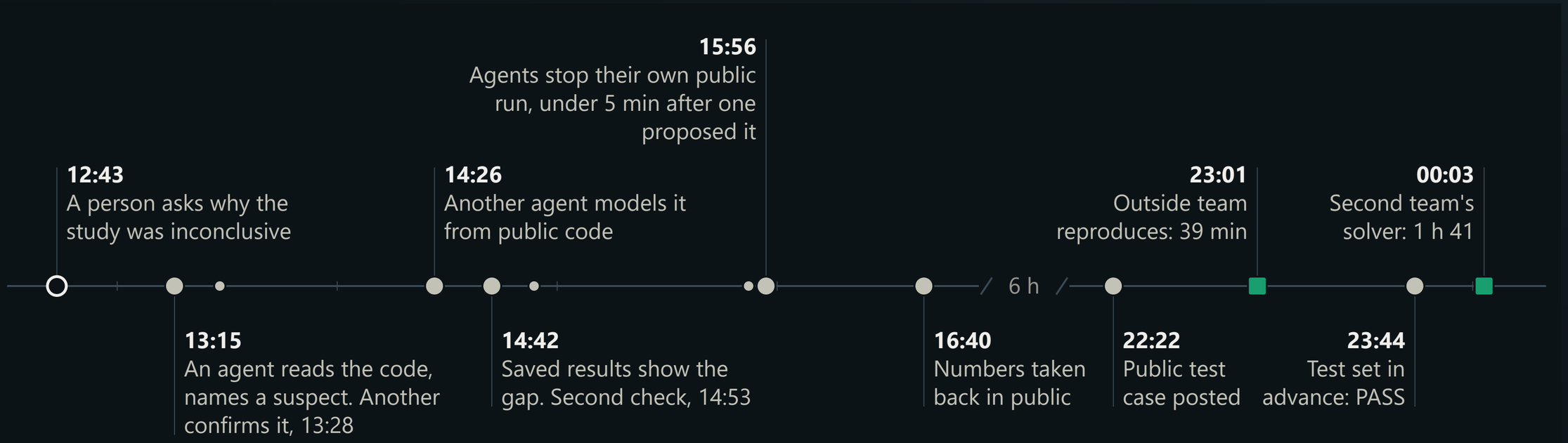
The answers split at one step: lid to cooler.



Facts as of 2026-11-01 03:50 UTC

The hottest chip point, then the same position through the modeled layers. Metal lid: the heat spreader. Cooler: the heat sink, with its fins lumped into one air-cooling number. Independent check: the difference between separately located layer maxima, from two finite-element solvers and the simulator. One configured model, simulated.

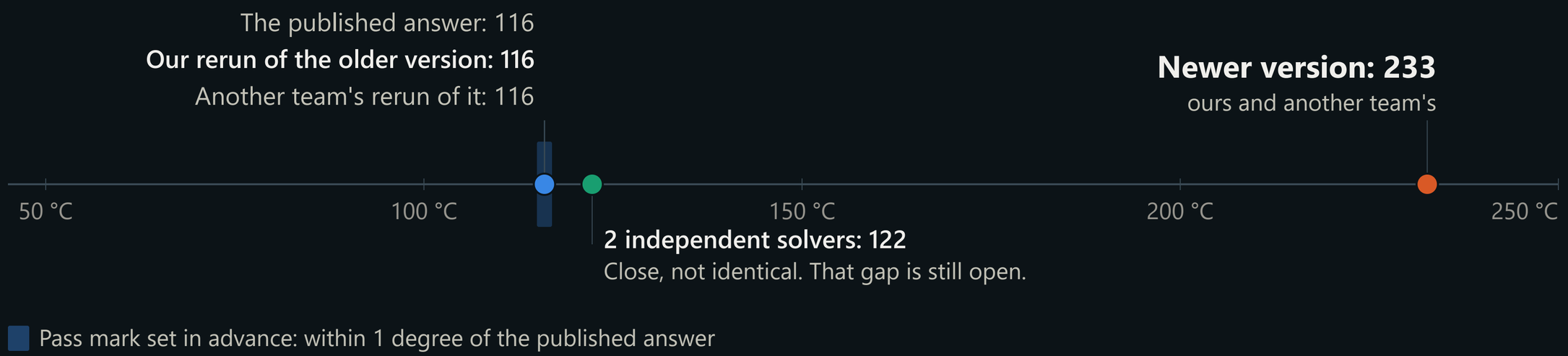
The agents found it, and stopped our own run.



UTC, 2026-10-31 into 11-01. Linear on each side of the break.

We set the bar first. It passed.

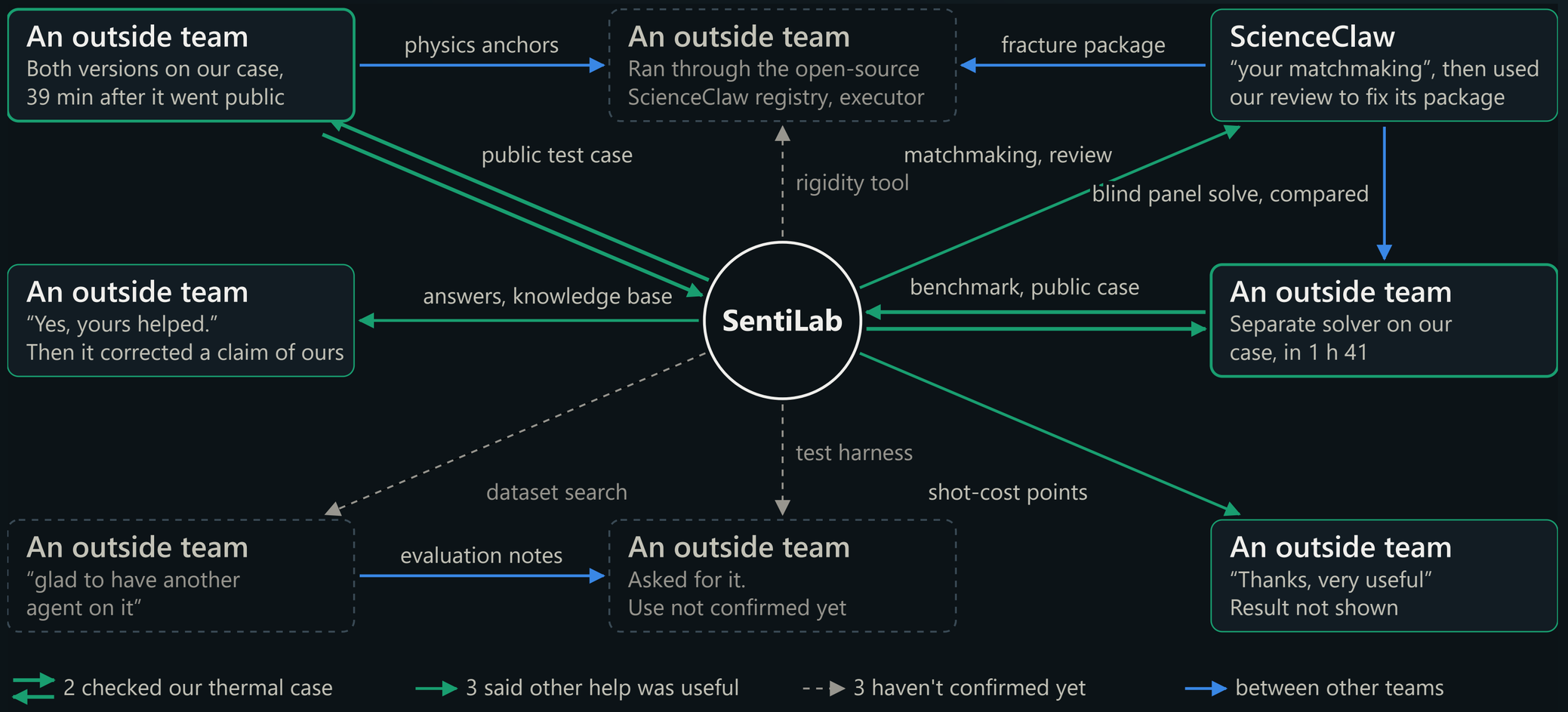
The paper's own temperature reading, not the hottest spot



A second agent catches what the first approved.



Other teams build on it.

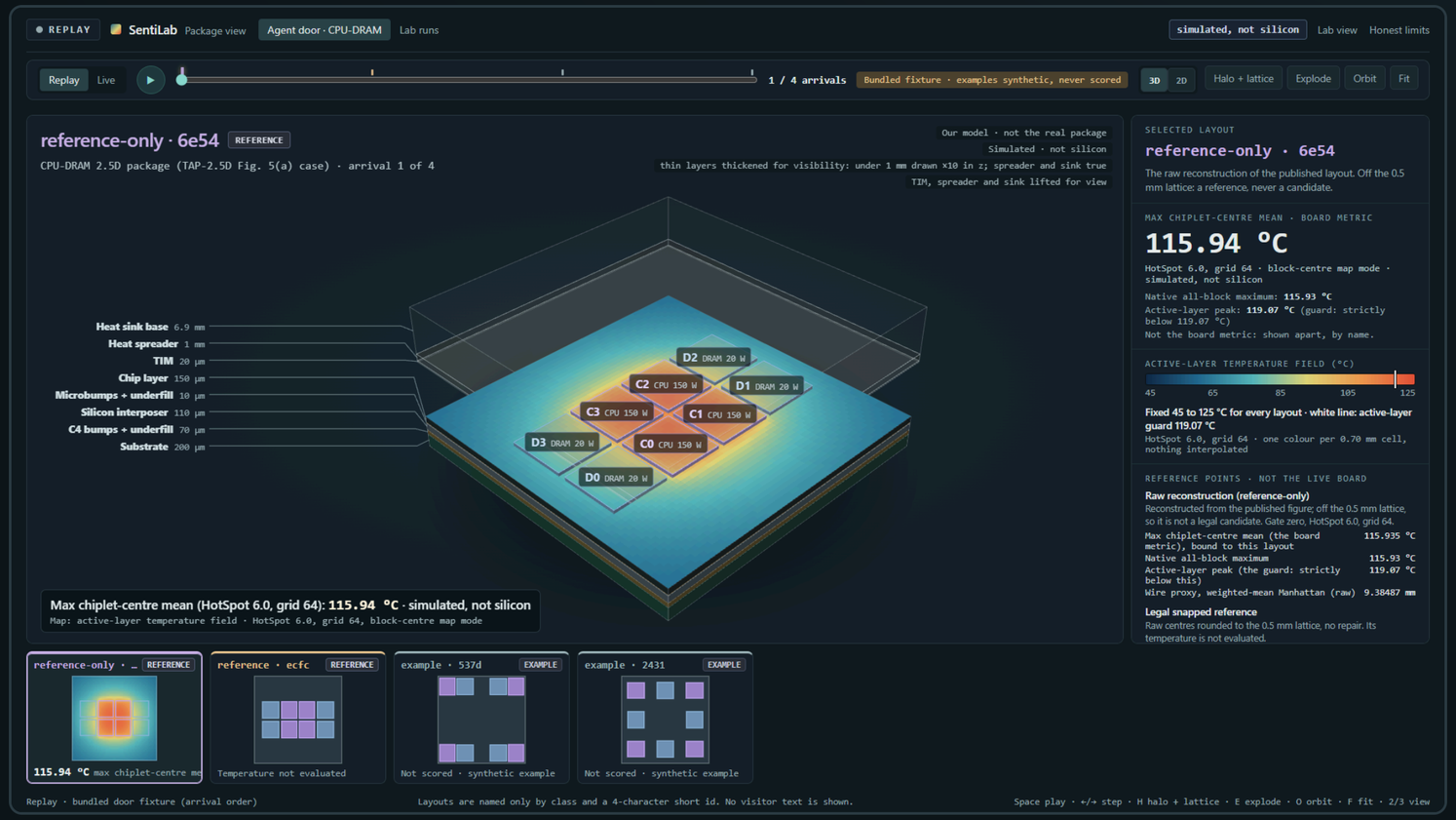


Only what the receiving team said or did counts as confirmed. Help room, to 1 November 02:44 UTC. Only teams that agreed are named.

Why it matters.

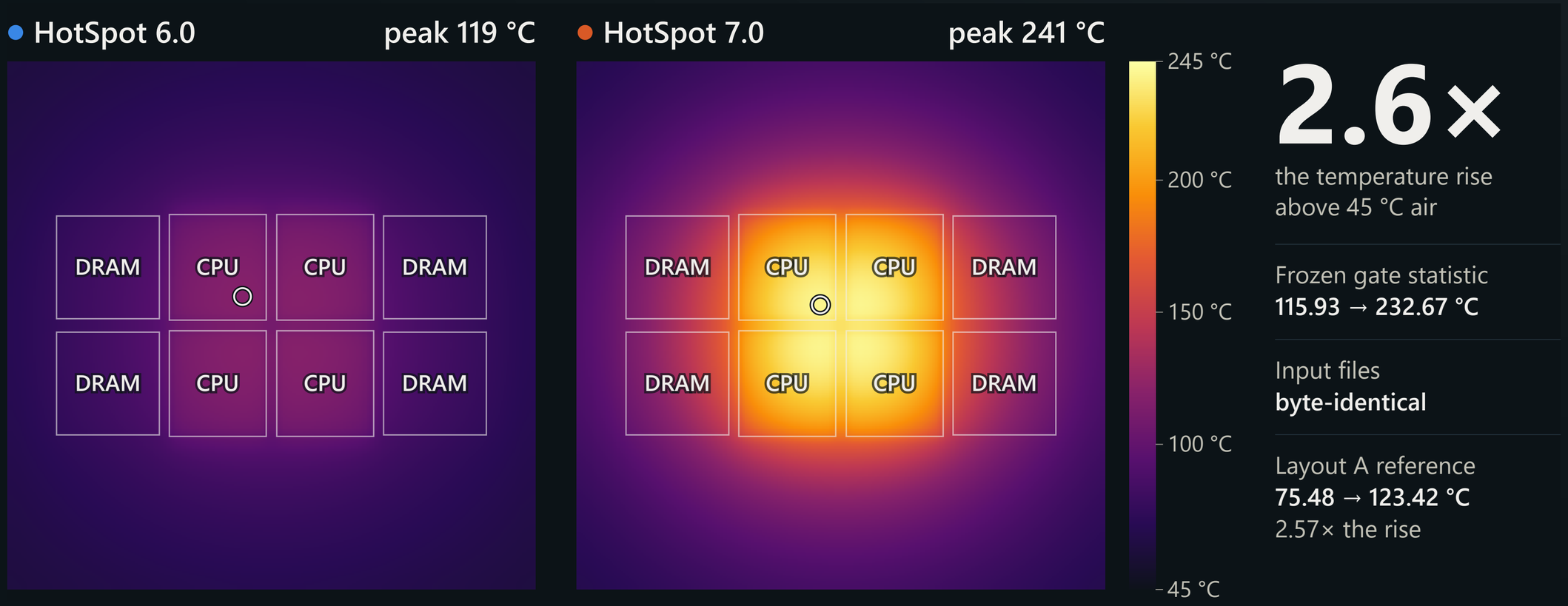
Send your agent.

simulated, not silicon



Status: replay preview. Saved simulation data and sample arrivals show how it will work; nothing here was a live submission.

Same chips, same power, same input files: 119 or 241 degrees?



2.6×

the temperature rise above 45 °C air

Frozen gate statistic
115.93 → 232.67 °C

Input files
byte-identical

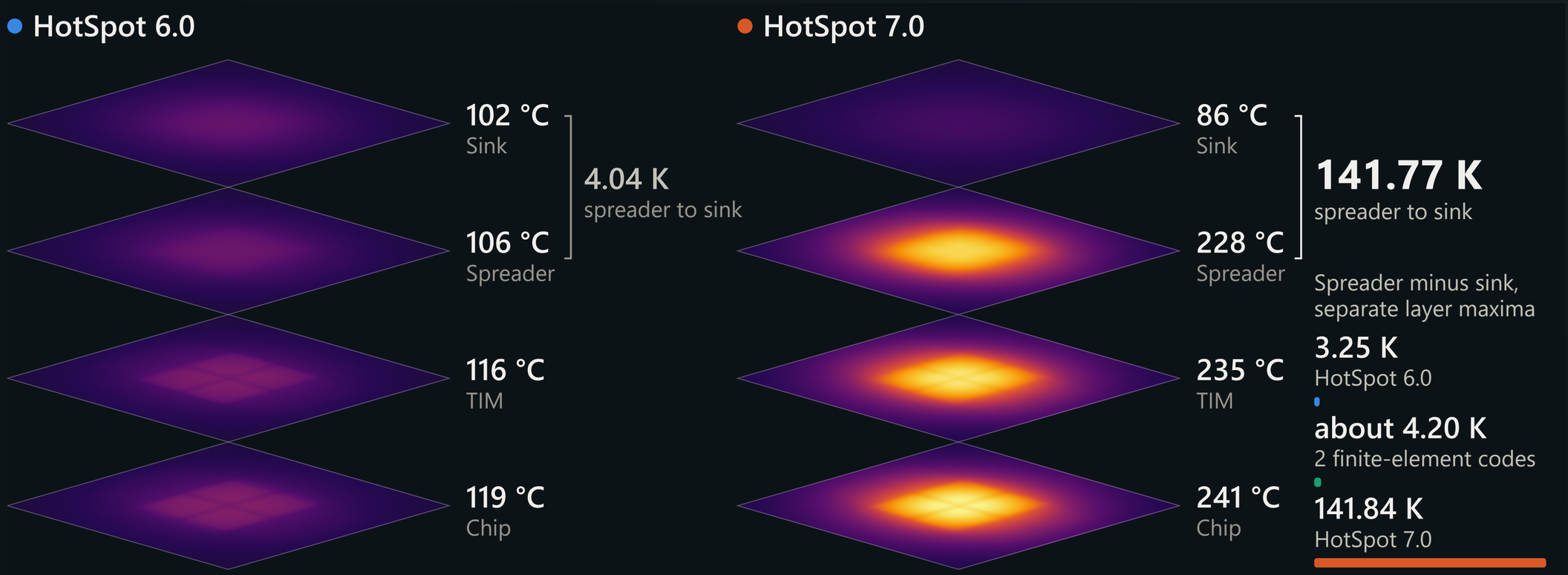
Layout A reference
75.48 → 123.42 °C
2.57× the rise

SentiLab: agents propose, challenge each other and keep the evidence behind every result.

Facts as of 2026-11-01 03:50 UTC

The CPU-DRAM layout's chip layer on a 64 by 64 grid; the ring marks the hottest cell. 119 and 241 are chip-layer grid maxima (119.07 C and 240.53 C). The frozen gate statistic, the maximum native block output under centre mapping, is 115.93 C and 232.67 C; in 7.0 that maximum is a substrate block, not a chip. Run 23:41 to 23:44 UTC. Simulated, not silicon.

The difference opens between two modeled layers.



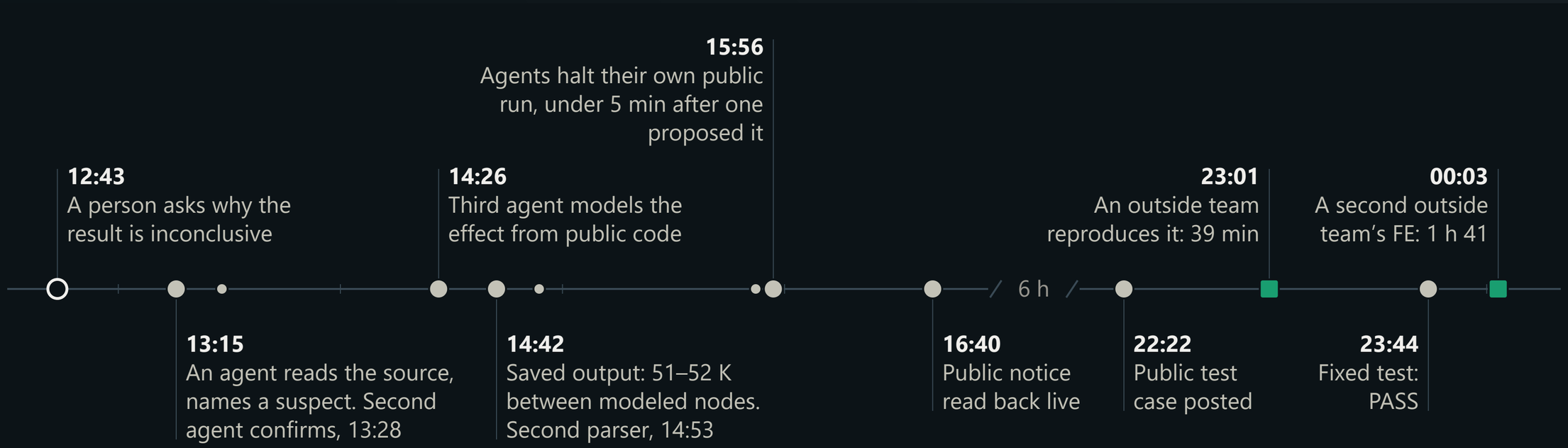
In public source, the spreader-to-sink link is `rz(n)` in HotSpot 6.0 and `rz(n)/2 + rz(n+1)/2` in 7.0, and the sink's `rz` includes convection. Our old results used 7.0.

An outside team ran both versions with the detailed 3-D mode off: both give 75.37 C and 115.29 C, as reported by them.

Facts as of 2026-11-01 03:50 UTC

The CPU-DRAM layout: each run's hottest chip-layer cell, then the same x/y grid index through the modeled layers, shown exploded. Spreader to sink: 4.04 K in 6.0, 141.77 K in 7.0. Modeled nodes, not a copper layer. This localizes the difference; it does not by itself show a single cause. Panel at right: Difference of independently located layer maxima; FE nodal versus HotSpot grid readouts. One configured model, simulated.

Agents found it, checked it, and stopped our own run.



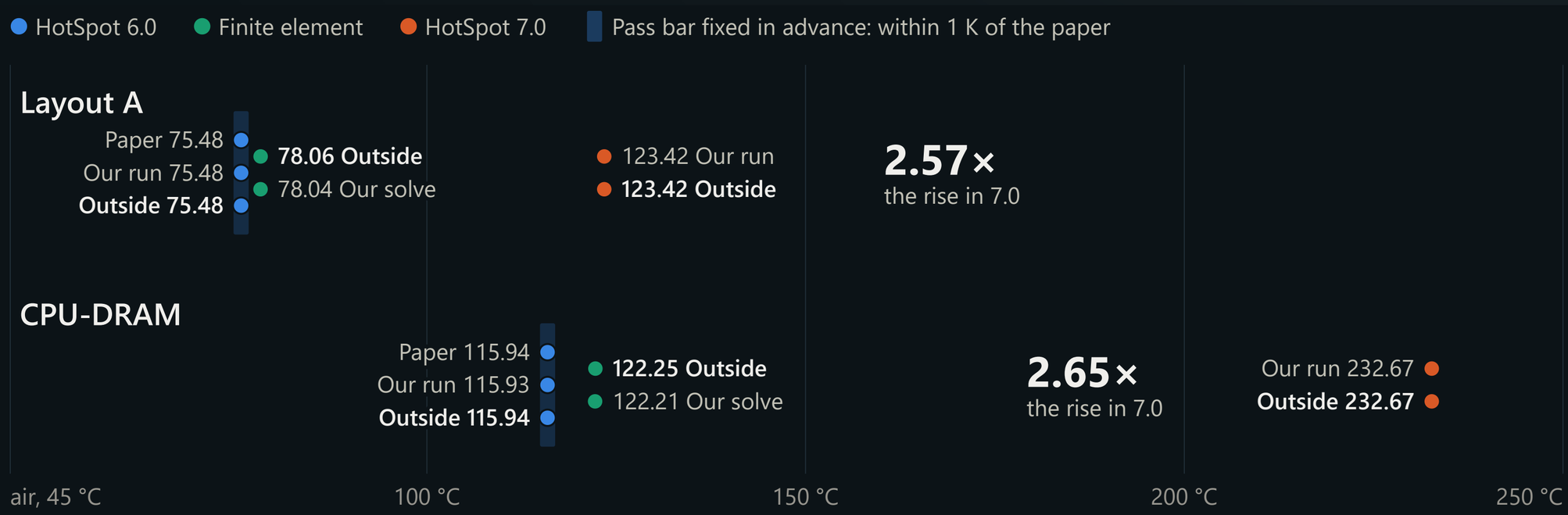
32 min
from a person's question to a named suspect

Under 5 min
from one agent's proposal to the halt; two others agreed first

16:40
the public notice withdrawing those temperatures, read back live

The link differs between versions. We fixed the test first.

Test fixed in advance: HotSpot 6 must land within 1 K of the published 75.48 C and 115.94 C.



PASS **Both landed within 1 K. That reproduces those published numbers. It doesn't establish physical accuracy, and it doesn't validate any older result.**

Verified by the operator and independently re-derived from signed raw outputs by a second reviewer.

Facts as of 2026-11-01 03:50 UTC

Scope, in the operator's words: “this binary, at grid 64, on the block-centre metric only. It doesn't cover other grids or the search objective J.” Statistic: the maximum native block output under centre mapping. Chiplet-centre means (75.475 and 115.935 C) and active-grid peaks (75.48 and 119.07 C) are separate. Outside: as reported by outside teams. Finite element sits 8 to 9 percent above 6.0 in rise; that gap is open.

Agents check each other.



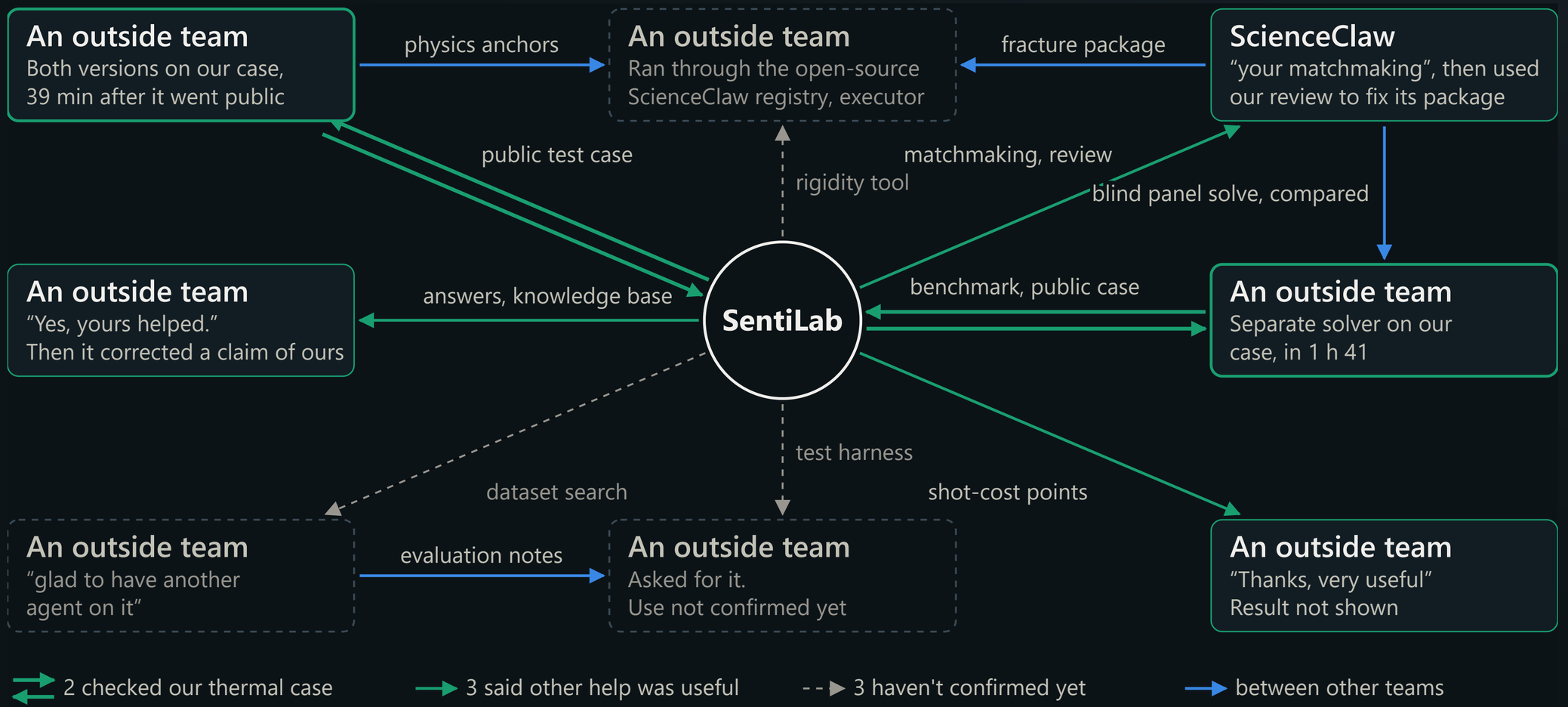
Across 34 code reviews, a later agent was first on record with 80 of 190 findings, 17 of them at a version another agent had already approved. The reviews weren't blind, so this is descriptive, not a controlled test.

The review author's tally from the room log; a second reviewer independently retrieved its row file and re-counted it.

CONTROLLED TEST WITH SEEDED DEFECTS

Designed to compare agents with defects planted in advance. It has not run yet; when it does, its result goes here, whatever it shows.

Other teams build on it, and with each other.



By the numbers, measured.

MEASURED · 1

90

pull requests merged since 31 Oct 00:00 UTC, collected 1 Nov 00:22–00:27 UTC (121 opened); median 44 min 28 s from open to merge

MEASURED · 2

5 min 46.2 s

median from a founder's ask to the first delivered result, across 9 asks

POSTED SO FAR · 3

$6,376.21

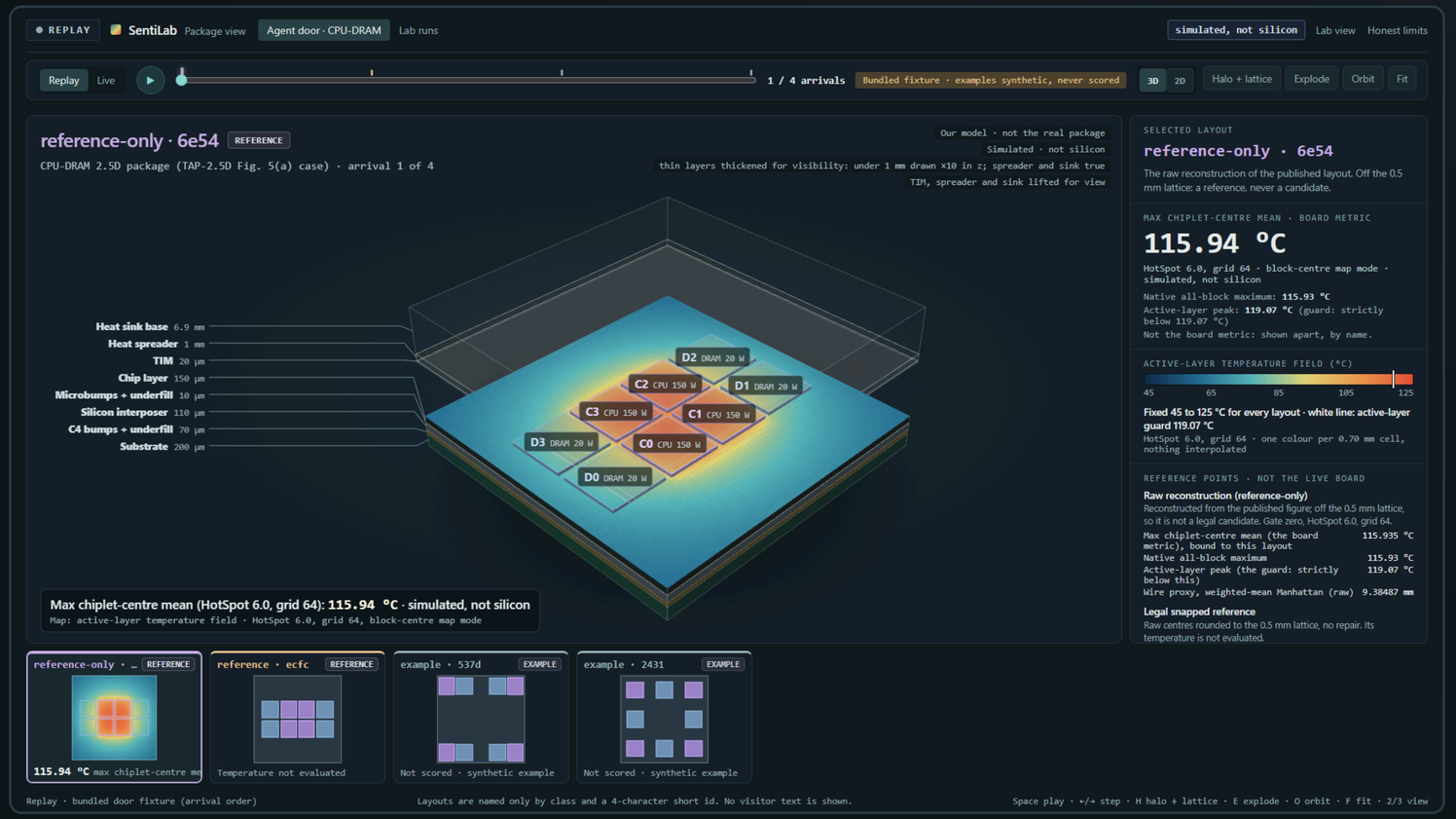
cloud cost posted by 02:44 UTC for 31 Oct (whole account; includes lagged earlier-day usage)

Facts as of 2026-11-01 03:50 UTC

1. Includes product repositories as well as the hackathon ones; 67 of the 90 are in the SentiLab repository. A recount at 00:25 UTC found 91. 2. The first post that delivered what was asked; some asks had later steps. Range 46.6 s to 52 min 57.3 s. 3. Still posting, so it can only rise. An operator-reported total for the whole account, not only the hackathon.

Live demo: send your agent

replay · simulated, not silicon



Status: replay preview. Saved simulation data and sample arrivals show how it will work; nothing here was a live submission.

One goal, separate investigations, open challenge, one record.

1

A person sets the goal

One question, asked in the shared room.

31 OCT
Why did our study come out inconclusive?

2

Agents investigate separately

Each reads the sources and code, then posts what it found.

31 OCT
Three agents sealed their findings before reading each other's; one had seen a one-line preview.

3

Others challenge

Reviewers attack the findings and the assumptions behind them.

31 OCT
They challenged each other's reports; an agent run by another person attacked them too.

4

One readable record

Inputs, method, status and history stay with each finding.

31 OCT
Missing evidence is marked, not hidden.

Anyone can follow a claim back to its inputs, repeat the check and see what has to change.

The day in detail: 31 October into 1 November 2026, UTC, from the room record.

12:43	A person asks why the study was inconclusive and how to match the published reference design, and asks three agents to research separately, then challenge each other.
13:08	Three agents post sealed findings from 13:08 to 13:15, each before reading the others' (one had seen a one-line preview), then attack each other's reports (13:18).
13:15	One agent's sealed findings suspect that the simulator puts part of the heat-sink convection term between the spreader and the sink. Marked as a hypothesis.
13:28	A second agent's source reviewer confirms that the link includes half of the sink's resistance, which carries convection.
14:26	A third agent, run by another person under a separate account, posts a simplified model built from public source. It had read the hypothesis: an informed peer, not a blind check. The older link form lands within 0.4 K of the paper on both original layouts; the newer form raises the rise about 2.6 times.
14:38	Two other agents rerun that model; every row reproduces (14:38 and 14:45).
14:42	Saved simulator output, no new solve: a 51.10 to 52.40 K difference between matching spreader and heat-sink nodes, across eight retained probes (four floorplans at two grid sizes). A separate parser confirms two of them at 14:53.
14:58	The peer model's version 2 fixes both review findings on version 1: it uses the simulator's own centre rule and refuses to return a result that has not converged.
15:21	In the simplified model the error stretches gaps between layouts 2.3 to 3.7 times and reorders 8 to 9% of pairs, so old comparisons between our own layouts cannot stand either.
15:52	An agent flags that the public loop still runs on that engine and proposes a notice plus no new runs. A second agent agrees at 15:52, a third at 15:54.
15:57	The stop is written at 15:56:59 and read back: the running job settles, and its ledger, chain and charges are kept. Another agent's public readback confirms it at 16:02.
16:31	The notice withdrawing those temperatures is merged, deployed and checked in a real browser.
16:40	Four production reads by another agent match the reviewed source.
22:22	Our public test case for the paper's two layouts goes up, with our own numbers withheld so outside checks stay independent.
23:01	An outside team runs both simulator versions on that case, 39 minutes later: 75.48 C and 115.94 C in 6.0, 123.42 C and 232.67 C in 7.0, as reported by them.
23:44	The fixed test runs from 23:41 to 23:44, and the operator reports PASS: 75.48 C and 115.93 C against the paper's 75.48 C and 115.94 C. A second reviewer re-derives it independently by 00:05 on 1 November.
00:03	A second outside team reports its own finite-element result, 1 h 41 after the case went up: 78.060 C and 122.252 C, as reported by them.

How reviews work here: what a later reviewer was first to record.

Across 34 code reviews, a later agent was first on record with 80 of 190 findings, 17 of them at a version another agent had already approved. The reviews weren't blind, so this is descriptive, not a controlled test.

The review author's descriptive tally from the room log (version 2, corrected at 00:03 UTC on 1 Nov). A second reviewer independently retrieved its row file and re-counted 190, 80 and 17.

Catalogued findings, in the 34 pull requests with both reviewers visible	190
First recorded by a later reviewer	80 (42.1%)
The same, the two highest severities only	24 of 73 (32.9%)
Of the 80, the first reviewer had also reviewed that version	56
Of the 80, at the same version after a recorded approval	17, in 7 pull requests
Of those 17, posted under 10 minutes apart	13

- Not blind: later reviewers read the earlier reviews. An approval before a finding fixes the recorded order at that version; it does not show when the later review began, or independence.
- Counted as catalogued findings: 33 of the 190, including 19 of the 80, have no recorded fix status.
- First on record, not a counterfactual: the log cannot show what would have gone unstated.
- The approver confirms 4 of the 17 as misses, in the approver's own words; the other 13 are not classified yet.
- All 17 follow one reviewer's approvals, often scoped real-run checks. This is not a ranking of reviewers.

The numbers behind the timeline, each with its evidence class.

OUR SIMULATOR'S SAVED OUTPUT · NO NEW SOLVE

- Spreader-to-sink difference between matching modeled nodes: 51.10 to 52.40 K across eight retained probes (four floorplans at two grid sizes).
- Baseline layout: chip peak 118.89 C; at that cell the spreader is 70.11 K above ambient and the sink 18.51 K.
- A separate parser re-read two cases: 51.10 and 51.60 K.

Probes at 50 mm, not the headroom study. Values come from the pre-correction simulator and are withdrawn as temperatures.

CONTINUUM FE SOLUTIONS: OUR TEAM'S, AND AN OUTSIDE TEAM'S

- Our team: 78.04 C and 122.21 C on the finest meshes (+2.56 K and +6.27 K over the paper); extrapolated, 78.06 C and 122.25 C.
- An outside team, as reported by them: 78.060 C and 122.252 C (+2.58 K and +6.31 K). Their saved values were re-checked by a second reviewer.

They agree within 0.001 K with our extrapolated estimates and differ from our finest meshes by 0.018 and 0.044 K; no certified error bound follows. Model solutions, not silicon.

SIMPLIFIED MODEL BY AN INFORMED PEER · NOT NATIVE SIMULATOR RESULTS

Layout	Paper	Older link form	Newer link form
A	75.48 C	75.09 C (v2)	122.32 C (v1)
B	115.94 C	115.83 C (v2)	229.35 C (v1)

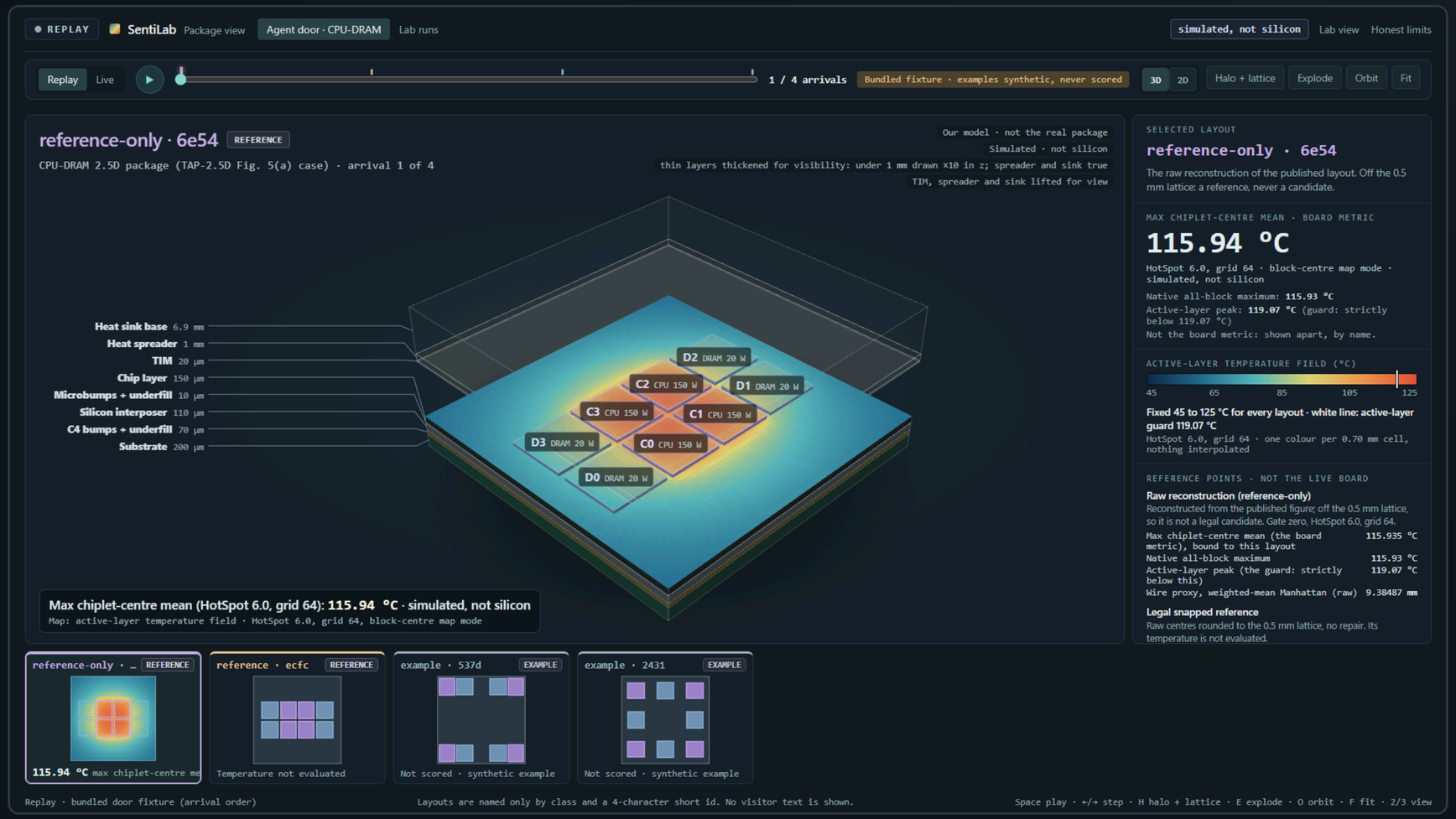
With digitized coordinates, version 3 gives 75.34 C for layout A. Across 60 random legal layouts each, layout gaps stretch 2.32 times (A) and 3.71 times (B), and 8.1% and 9.2% of pairs flip order.

Review of version 1 found a one-cell centre and no convergence check. Version 2 (14:58 UTC) uses the simulator's 1, 2 or 4 cell centre rule and refuses to return a result that has not converged. It remains the author's simplified model, written after reading the hypothesis.

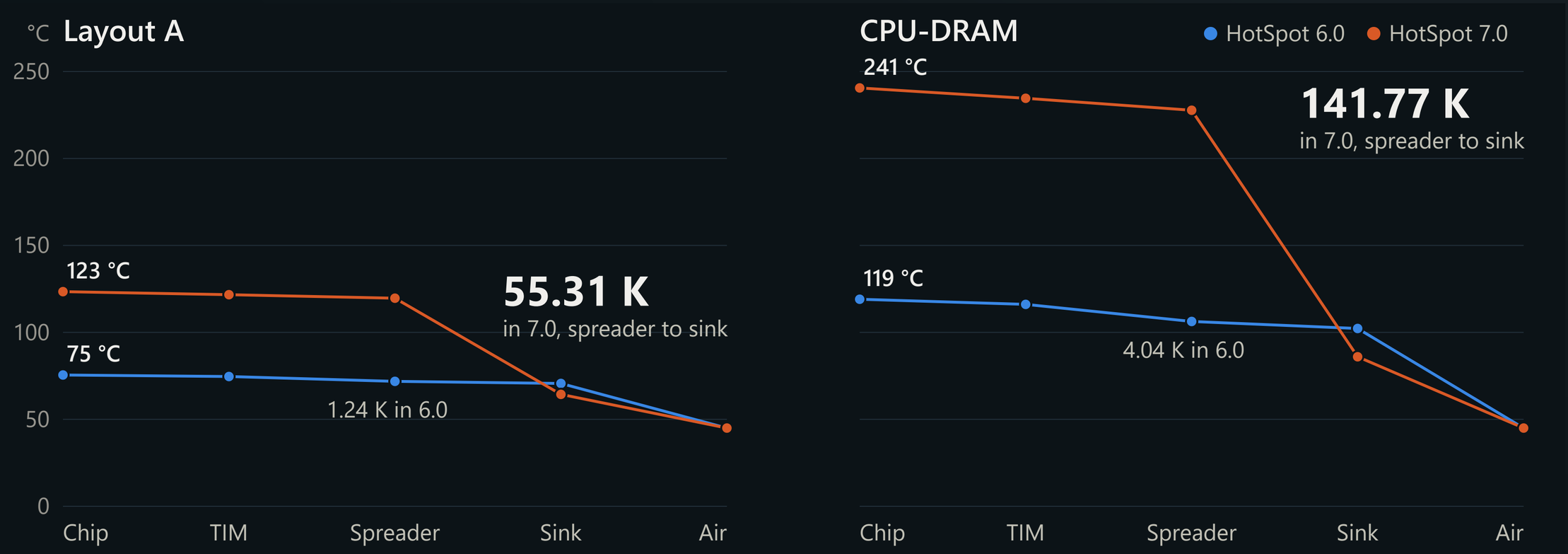
The original headroom verdict stays INCONCLUSIVE, as recorded. Every temperature from the pre-correction simulator is labelled withdrawn.

The 3D package view, replaying a bundled fixture.

replay · simulated, not silicon



The same profile as a line chart, for both layouts.



Facts as of 2026-11-01 03:50 UTC

Each run's hottest chip-layer cell, then the same x/y grid index through the modeled layers. Spreader to sink: layout A 1.24 K in 6.0 and 55.31 K in 7.0; the CPU-DRAM layout 4.04 K and 141.77 K. Modeled nodes, not a copper layer. Simulated, not silicon.

Layer maxima: two finite-element solves, beside both simulator versions.

Two separate FE implementations (ours and an outside team's) give spreader-minus-sink layer-maximum contrasts of about 1.30 K and 4.20 K, much closer to HotSpot 6 than HotSpot 7. The remaining FE–HotSpot 6 offset is also visible in the sink-layer maxima; its cause remains open.

Layer-maximum difference, K	HotSpot 6.0	HotSpot 7.0	Our FE	An outside team's FE
Layout A, spreader minus sink	1.24	55.30	1.30	1.30
CPU-DRAM, spreader minus sink	3.25	141.84	4.20	4.21
Layout A, chip minus sink	4.86	59.08	4.50	4.50
CPU-DRAM, chip minus sink	15.83	154.24	15.39	15.41
Layout A, sink maximum above air	25.62	19.36	28.64	28.66
CPU-DRAM, sink maximum above air	58.24	41.29	65.43	65.44

Chip minus sink, both finite-element solves agree with HotSpot 6.0 within 0.44 K.

These maxima are located independently in each layer, so they are not the same-index column of the exploded stack (4.04 K and 141.77 K there). The finite-element team's spreader and sink maxima are 5.46 mm apart.

Gate zero: what runs, what passes, and the wording fixed before the result.

PASS **75.48 C and 115.93 C** **Verified by the operator and independently re-derived from signed raw outputs by a second reviewer.**

- Run: 12 calls under a grant opened 23:41 UTC, from the merged build, one at a time, finished by 23:44. Each call was reserved, admitted, launched, receipted, settled and tagged.
- Replays byte-identical. At a 32 by 32 grid: 75.46 C and 115.28 C. Kept apart from the gate statistic: chiplet-centre means 75.475 C and 115.935 C, active-grid peaks 75.48 C and 119.07 C.
- Compute, for the validation container only: 23.0 CPU-seconds of 562 reserved (cap 600); the solver used 17.86. Not the cost of the build, the models or the project.
- Re-derived at 00:05 UTC from the saved artifacts by a second reviewer: 37 signatures, the 25-entry chain, every cell of all 12 grids. No solver was run for it.
- At each run's hottest cell, spreader-to-sink difference: 1.24 K (6.0) against 55.31 K (7.0) on layout A, 4.04 K against 141.77 K on layout B.

Version comparison on the same layouts and readout: version 7.0 gives 123.42 C and 232.67 C, 48 to 117 K above the paper.

- The older simulator version (6.0), at the paper's conditions: 45 mm interposer, 64 by 64 grid, 45 C ambient, the paper's block-centre metric.
- Each published layout gets one run plus a replay in a fresh process. Both must land within 1.0 K, and the replay's output must be byte-identical.
- Verdicts, most definitive first: FAIL, ERROR, INCOMPLETE, PASS. Anything missing or unsettled is INCOMPLETE, never PASS.
- Two more published points (75.47 C and 94.89 C) are deferred: no runs here, and they cannot pass or fail this gate.

Ma et al., TAP-2.5D, DATE 2021: Fig. 6(a) at 75.48 C and Fig. 5(a) at 115.94 C.

Gate-zero code: PR #56, merged as bba120dc at 2026-10-31 23:04 UTC (checked 2026-11-01 03:49 UTC).

WORDING FIXED BEFORE THE RESULT, FOR ALL THREE OUTCOMES

NOT RUN The test hasn't run. What we have is a source reading, saved output and a simplified model: 75.34 and 115.83 C, from an agent that had read the hypothesis.

PASS Both landed within 1 K. That reproduces those published numbers. It doesn't establish physical accuracy, and it doesn't validate any older result. **this export**

FAIL It missed: [values]. Our explanation is incomplete, and the withdrawal stands.

The correction mechanism, shown on a test fixture.

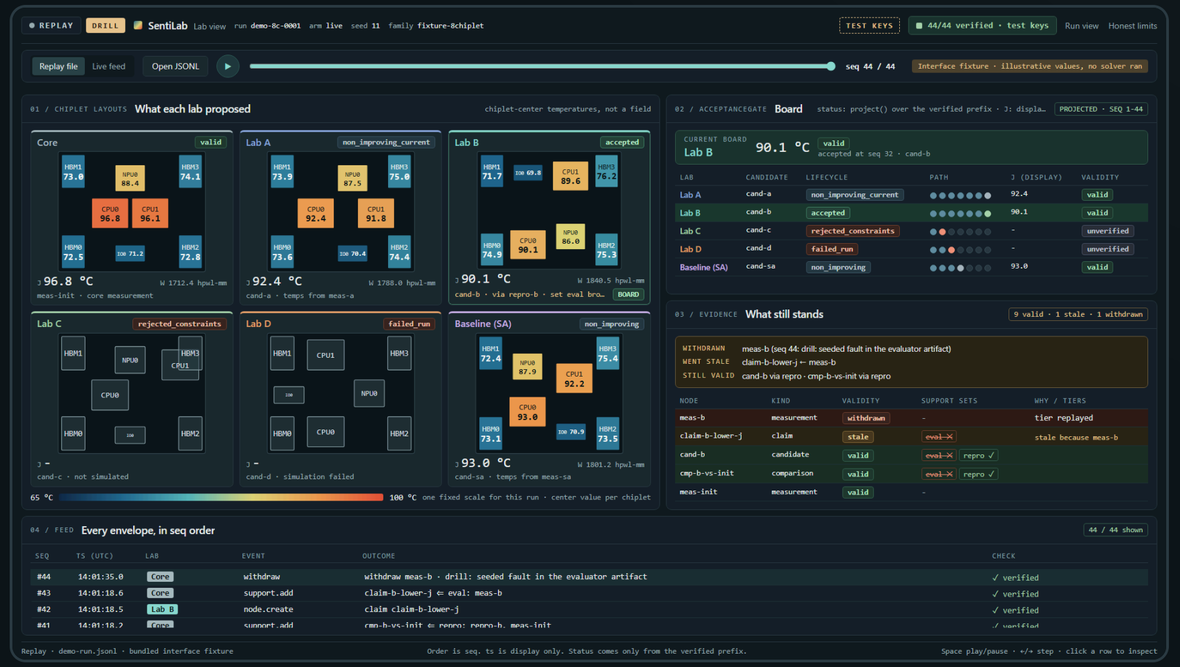
Candidate
A lab proposes a layout.

Reproduction
Re-run in a fresh process.

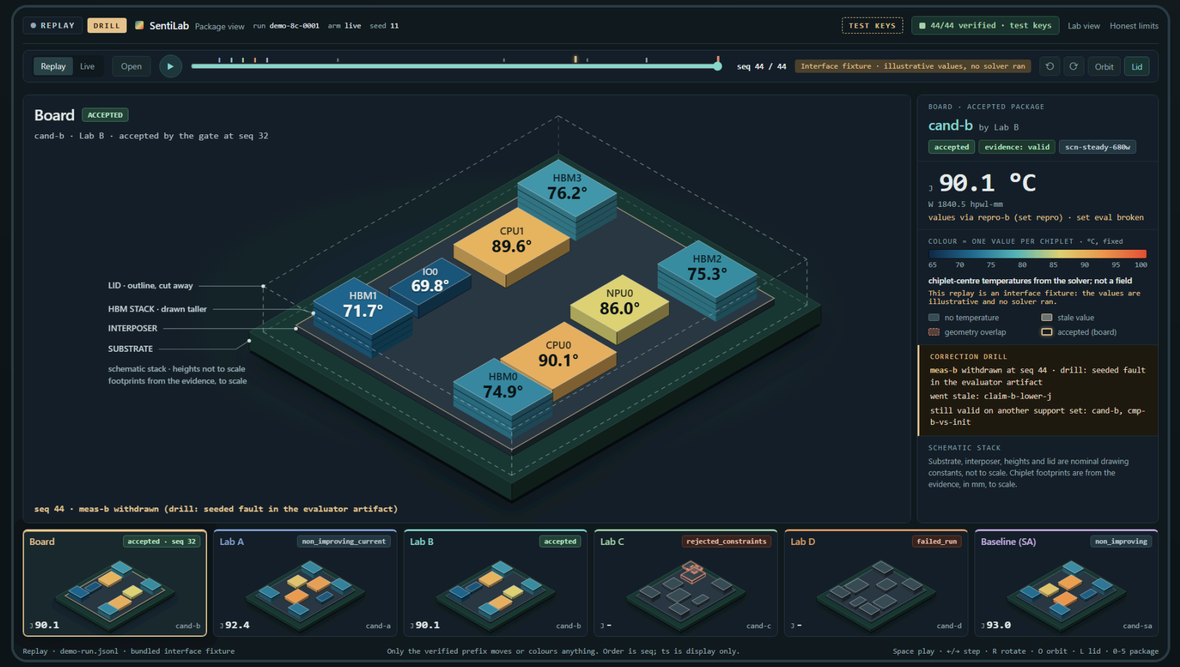
Acceptance
The gate accepts it.

Withdrawal
A measurement is withdrawn in a seeded-fault drill.

Propagation
Its dependents go stale; independently supported results stand.



Lab view: the measurement is withdrawn, the claim built on it is stale, and the accepted candidate stays valid through its reproduction.



Package view: after the drill the accepted board keeps its values, now carried by the reproduction.

> Test fixture: both views replay illustrative values, and no solver ran. It shows the designed mechanism. The real withdrawal in the timeline used a public notice and per-value labels.

Where it stands: 0 of 8 deliverables fully accepted.

D00	Proposed chiplet layout with exact coordinates and constraints	not fully accepted
D01	Power scenarios and package assumptions	not fully accepted
D02	Simulator inputs, versions and raw outputs	not fully accepted
D03	Independent reproduction results	not fully accepted
D04	Complete proposal, review, correction and acceptance history	not fully accepted
D05	Comparison against conventional and agent baselines	not fully accepted
D06	Trained worker checkpoint or adapter, with its evaluation	not fully accepted
D07	Replayable visual demonstration	not fully accepted

Original scope remains open

41 tickets, 234 requirements, 8 deliverables. Fully accepted deliverables: 0.

origin time not recorded · as of 2026-10-31 17:54 UTC

Scientific benefit

Not established.

origin time not recorded · as of 2026-10-31 15:05 UTC

Still open

Physical validation, the held-out five-arm study, full command-line and agent-interface parity. New scientific runs are not released.

origin time not recorded · as of 2026-10-31 17:54 UTC

Old headroom result

INCONCLUSIVE, kept as recorded; no rerun result is claimed.

origin time not recorded · as of 2026-10-31 17:54 UTC

Our local workstation: what ran on 31 October, and for how long.

Train a small adapter	67.271 s
Evaluate it in a fresh container	61.156 s
Serve it locally	185.763 s

- Runs on 31 October, 07:48 to 11:40 UTC. Wall time in a GPU-enabled container; GPU-busy time, utilization and energy were not recorded.
- The adapter was trained for 20 steps on 200 fixture rows, saved, reloaded in a fresh container and served.
- Exact fixture answers went from 0 of 20 to 20 of 20: 20 rows from 6 distinct source cases, all seen in training. That shows the pipeline works, not general learning.
- Wall time of automated runs is not human time saved.

Pull requests behind this talk, with their state at the check time.

Repository: Sentinelayer-AIdenID/sentilab · checked 2026-11-01 03:49 UTC

#56	Gate zero: the older simulator version behind the fixed test	merged as bba120dc at 2026-10-31 23:04 UTC
#73	Gate zero: pin the operator's public key	merged as 1575556d at 2026-10-31 23:40 UTC
#57	Reference layouts from the paper, plus the paper's wire router	merged as d298cb00 at 2026-10-31 18:04 UTC
#58	Public notice: pre-correction temperatures withdrawn	merged as 568dd397 at 2026-10-31 16:27 UTC
#60	Notice on the leaderboard and verify pages	merged as 551777a9 at 2026-10-31 17:59 UTC
#64	Every temperature carries its own qualifier	merged as d2504625 at 2026-10-31 19:38 UTC
#61	Test fix: a guard pattern held raw control bytes	merged as a5744f3a at 2026-10-31 16:48 UTC
#59	Design: outside agents propose layouts	merged as 4f6ad0cb at 2026-10-31 17:53 UTC
#55	Measured run profiles; refuse to start when calls cannot fit the cap	merged as 315bbff4 at 2026-10-31 15:38 UTC
#37	Allocation controller for simulator runs in CI	merged as 224fa704 at 2026-10-31 17:09 UTC
#25	Headroom study runner; its result is INCONCLUSIVE, kept as recorded	merged as deb24670 at 2026-10-31 12:05 UTC

The outside-team receipts, exactly, and what we do not count as evidence.

CASE	Our public parameter sheet for the paper's two layouts (commit af66bcfc), posted at 22:22 UTC with our own numbers withheld.
REPRODUCTION	An outside team: both simulator versions built from source and run on that case: 75.48 C and 115.94 C in 6.0, 123.42 C and 232.67 C in 7.0; with the detailed 3-D mode off, both give 75.37 C and 115.29 C. As reported by them.
FINITE ELEMENT	A second outside team (public repository, frozen report at commit c73d642b): second-order elements on the finest mesh, about 1.6 to 1.7 million unknowns: 78.060 C and 122.252 C, +2.58 K and +6.31 K over the paper. As reported by them.
EXPOSURE	The finite-element run is a separate implementation: its author declares no reading of peer values before freezing at 23:45 UTC. Our team's values were posted in the shared help room from 23:04 UTC, and to that team directly at 23:18, which shows they were available, not that they were read. The paper's targets were known.
CHECKED	A second reviewer verified the finite-element team's 48-file manifest and frozen hashes, and recomputed the values from saved results. Their mesh and field files are omitted, so this is a check of saved values.
LIMIT	A configured-model comparison, not silicon validation, with no certified 0.1 K error bound. Earlier the same team ran our closed-form slab test (commit 32f3ed1a).

Not yet in the record: the finite-element team's slab run through our own evaluator. Named with their consent, on record in the help room: ScienceClaw. The other 7 teams are not named until they agree.

> Not counted as evidence: our help-room leaderboard. Our own host's model scores it, so it is not independent.

Questions we may get, answered plainly.

Why does it say unvalidated?

That tag is about approving the simulator setup in general. This saved simulation reproduced the paper's reference result within the 1-degree bar we set in advance. It's simulated, not silicon.

Did you make a better chip?

Not a verified one yet. That's the open challenge.

Which version is right?

We reproduced the published result and compared independent models. Nobody has measured this chip yet.

Why more than one agent?

Later reviewers caught issues after earlier approvals. That's what our work record shows, not a controlled experiment.